① Interp. polynom. Kurve zu

t_i	0	1	2	3
z_i	$\begin{pmatrix}0\\0\end{pmatrix}$	$\begin{pmatrix}1\\1\end{pmatrix}$	$\begin{pmatrix}1\\0\end{pmatrix}$	$\begin{pmatrix}0\\1\end{pmatrix}$

$$\left.\begin{array}{l}\underline{t_i,\ z_{i,1}}\\ \underline{t_i,\ z_{i,2}}\end{array}\right\} \text{ IP nach Newton (dividierte Diff.)}$$

$$\begin{array}{llllll}
t_i & z_i & & & & \\
0 & \begin{pmatrix}0\\0\end{pmatrix} & & & & \\
1 & \begin{pmatrix}1\\1\end{pmatrix} & \dfrac{\begin{pmatrix}1\\1\end{pmatrix}-\begin{pmatrix}0\\0\end{pmatrix}}{1-0}=\begin{pmatrix}1\\1\end{pmatrix} & & & \\
2 & \begin{pmatrix}1\\0\end{pmatrix} & \dfrac{\begin{pmatrix}1\\0\end{pmatrix}-\begin{pmatrix}1\\1\end{pmatrix}}{2-1}=\begin{pmatrix}0\\-1\end{pmatrix} & \dfrac{\begin{pmatrix}0\\-1\end{pmatrix}-\begin{pmatrix}1\\1\end{pmatrix}}{2-0}=\begin{pmatrix}-\frac{1}{2}\\-1\end{pmatrix} & & \\
3 & \begin{pmatrix}0\\1\end{pmatrix} & \dfrac{\begin{pmatrix}0\\1\end{pmatrix}-\begin{pmatrix}1\\0\end{pmatrix}}{3-2}=\begin{pmatrix}-1\\1\end{pmatrix} & \dfrac{\begin{pmatrix}-1\\1\end{pmatrix}-\begin{pmatrix}0\\-1\end{pmatrix}}{3-1}=\begin{pmatrix}-\frac{1}{2}\\1\end{pmatrix} & * &
\end{array}$$

$$* \quad \frac{\begin{pmatrix}-\frac{1}{2}\\1\end{pmatrix}-\begin{pmatrix}-\frac{1}{2}\\-1\end{pmatrix}}{3-0}=\begin{pmatrix}0\\\frac{2}{3}\end{pmatrix}$$

$$p(t)=\begin{pmatrix}0\\0\end{pmatrix}+\begin{pmatrix}1\\1\end{pmatrix}(t-0)$$
$$+\begin{pmatrix}-\frac{1}{2}\\-1\end{pmatrix}(t-0)(t-1)$$
$$+\begin{pmatrix}0\\\frac{2}{3}\end{pmatrix}(t-0)(t-1)(t-2)$$